

StrataOne

Field Installation Guide for Administrators

Enterprise preview administrator guide for installing, securing, operating, and validating StrataOne control plane deployments.

Field	Value
Product	StrataOne Infrastructure Orchestrator
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Audience	Platform administrators, deployment admins, security admins, and field installation engineers
Deployment mode	Enterprise preview / controlled lab or pilot rollout
Repository basis	Current local StrataOne repository after expanded deployment-template catalog

Admin rule

Do not enable live provider or live Redfish actions against production hardware until credentials, approval policy, rollback plan, OEM lab-validation evidence, and change-control approval are in place.

Table of Contents

Contents	3
1. Current Status for Admins	3
2. Reference Architecture	4
3. Installation Sequence	4
3.1 Prepare the Repository and Environment	4
3.2 Start and Verify the Control Plane	5
4. First-Run Administrator Configuration	6
5. Security and Governance Baseline.....	6
6. Deployment Templates.....	7
7. Standard Deployment Workflow	7
8. Hypervisor-Specific Admin Notes	8
Azure Local.....	8
VMware vSphere.....	8
Nutanix AHV	9
Proxmox VE	9
Hyper-V	9
KVM.....	9
OpenShift Virtualization.....	10
9. ISO Library and Virtual Media	10
10. Day-2 Operations	10
11. Backup and Recovery	11
12. Troubleshooting Quick Reference	11
13. Field Readiness Checklist	12

Contents

- Current implementation status and maturity boundaries
- Reference architecture and service ports
- Production-ready installation sequence
- Security, RBAC, secrets, approvals, and audit configuration
- Deployment templates and hypervisor-specific admin notes
- Operational workflows, backup, recovery, and troubleshooting

1. Current Status for Admins

Area	Status	Admin interpretation
Dashboard, auth, RBAC, sessions, logout	Implemented	Use named admin accounts. Change default passwords before production.
PostgreSQL state and Redis queue	Implemented	Compose defaults use PostgreSQL and Redis; SQLite remains local-dev fallback.
Deployment wizard and templates	Implemented	15 built-in templates plus browser-local template import/upload.
Validate, plan, inventory, preflight	Implemented	Inventory depends on reachable Redfish/BMC endpoints and credentials.
Artifact generation and ZIP download	Implemented	Includes SiteSpec, plan, reports, Terraform, Bicep, and Ansible handoff files.
Approvals and approval policies	Implemented	Protected actions can require roles, minimum vote count, and expiry.
Live Redfish virtual media	Guarded preview	Disabled by default; requires live flag and OEM validation evidence.
Azure Local deploy execution	Staged contract	Provides stages and artifacts; real field execution must be validated per environment.
vSphere/AHV/Proxmox/Hyper-V/KVM/OpenShift execution	Provider contract	Templates, plans, and handoff exist; full live execution requires provider-specific validation.

What this guide is

This is an administrator field-installation guide. It explains how to stand up and govern the StrataOne control plane and how to prepare deployments safely. It is not a claim that every vendor and hypervisor provider has completed OEM-by-OEM production certification.

2. Reference Architecture

Service	Purpose	Default port
strataone-api	FastAPI service for validation, orchestration, auth, approvals, providers, jobs, and artifacts	8080
strataone-dashboard	Nginx-hosted operator dashboard	8088
strataone-proxy	Optional single front door for dashboard/API routing	8089
strataone-worker	Redis-dispatched job worker for queued execution	internal
postgres	Durable state for sites, jobs, inventory, users, roles, approvals, audit, provider config	5432 internal
redis	Queue dispatch and worker coordination	6379 internal

- Use the dashboard for operator workflows and the API for automation, CI, and emergency administration.
- Use PostgreSQL and Redis for shared or multi-admin environments. Use SQLite only for isolated development.
- Persist runtime data under the repository `.strataone` directory when using the supplied Compose file.

3. Installation Sequence

Prerequisite	Admin action
Host	Windows, Linux, or VM capable of running Docker Engine / Docker Desktop and Docker Compose.
Ports	Reserve 8088 for dashboard, 8089 for proxy, and 8080 for API or override them in <code>.env</code> .
Credentials	Prepare admin password, bootstrap token, PostgreSQL password, and any named API tokens.
Secrets	Choose env, file, or Vault-compatible provider before onboarding production BMC credentials.
Network	Confirm admin workstation reachability to dashboard/API and BMC network path from worker/API.
Change control	Create approval policy before enabling live Redfish or provider actions.

3.1 Prepare the Repository and Environment

1. Clone or update the StrataOne repository on the installation host.
2. Create a `.env` file from `.env.example` or define environment variables in the shell/session manager.

3. Set STRATAONE_ENVIRONMENT=production for real installations.
4. Replace STRATAONE_ADMIN_PASSWORD, STRATAONE_BOOTSTRAP_TOKEN, POSTGRES_PASSWORD, and any placeholder secrets.
5. Set STRATAONE_CORS_ORIGINS to the exact dashboard/proxy origins. Do not use wildcard origins when auth is enabled.
6. Decide whether operators will use <http://host:8088> directly or the proxy/front-door at <http://host:8089>.

Item	Admin Guidance
STRATAONE_AUTH_ENABLED	true for any shared environment. false only for isolated local development.
STRATAONE_STATE_BACKEND	postgres for production; sqlite only for local development.
STRATAONE_QUEUE_BACKEND	redis for queued/distributed execution.
STRATAONE_REQUIRE_APPROVALS	true for live, ISO, node replacement, and deployment actions.
STRATAONE_ENABLE_LIVE_REDFISH	false until OEM lab validation and approval policy are complete.
STRATAONE_REQUIRE_OEM_VALIDATION	true when live provider actions are allowed only after validation evidence exists.
STRATAONE_ISO_LIBRARY_DIR	.strataone/iso-library by default; mounted in the API container as /app/.strataone/iso-library.

3.2 Start and Verify the Control Plane

Task	Command or target
Build and start	docker compose up -d --build
View containers	docker compose ps
API health	curl http://localhost:8080/health
Dashboard	Open http://localhost:8088 or proxy http://localhost:8089
Stop stack	docker compose down
Stop and remove volumes	docker compose down -v only when intentionally deleting state

Production password gate

When STRATAONE_ENVIRONMENT=production, startup should be treated as invalid if default placeholder bootstrap, admin, or database passwords remain configured.

4. First-Run Administrator Configuration

Admin task	Where / how
Sign in	Use the configured admin account. Change default password before any other operator is onboarded.
Create users	Settings > Access & RBAC. Assign roles from the dropdown rather than sharing admin credentials.
Review roles	Use Viewer, Operator, Deployment Admin, Platform Admin, and Auditor as starting points.
Configure secrets	Settings > Secrets. Prefer Vault or secret references over inline BMC passwords.
Configure providers	Settings > Providers and Providers page. Add, edit, validate, or mark provider evidence.
Set approvals	Settings > Approval policy. Require approvals for mount-iso, deploy-azure-local, and node-replacement.
Set ISO library	Create .strataone/iso-library on the host and place approved boot media there if using local browse.
Audit baseline	Check Settings > Audit after login, role, provider, and approval changes.

5. Security and Governance Baseline

Control	Admin requirement
Authentication	Keep auth enabled. Use password login for dashboard users and API tokens only for automation.
RBAC	Grant least privilege. Separate platform admins, deployment admins, operators, viewers, and auditors.
Session revoke	Use session management to revoke abandoned or suspicious sessions.
CORS	Allow only known dashboard/proxy origins. Wildcard origins are not production-safe.
Secrets	Store only references in deployment definitions. Do not commit BMC, vCenter, Prism, Azure, or Vault credentials.
Rate limiting	Keep enabled for exposed API endpoints. Tune request/window values for automation volume.
Security headers	Use the proxy or dashboard Nginx configs for CSP, frame protection, content sniffing protection, and referrer policy.
Tenant scope	Keep tenant enforcement enabled for multi-team or shared admin environments.

6. Deployment Templates

Template import

Admins and deployment leads can use the Deployments page to select built-in templates or import JSON/YAML deployment templates. Imported templates are browser-local unless later promoted into a shared GitOps or server-backed template workflow.

Template	Platform	Nodes	Use case
Azure Local Branch	azure-local	2	branch
Azure Local Edge Scale	azure-local	4	edge
Azure Local Stretched	azure-local	4	resilience
vSphere Management Cluster	vmware-vsphere	3	management
vSphere Workload Cluster	vmware-vsphere	4	private-cloud
Nutanix AHV Edge	nutanix-ahv	3	edge-hci
Nutanix AHV ROBO	nutanix-ahv	2	branch
Proxmox Lab	proxmox	2	lab
Proxmox Edge	proxmox	3	edge
Hyper-V Cluster	hyper-v	2	private-cloud
Hyper-V Workload Scale	hyper-v	4	private-cloud
KVM Libvirt Cluster	kvm	3	private-cloud
KVM OpenStack Compute	kvm	4	cloud-compute
OpenShift Virtualization	openshift-virtualization	3	private-cloud
OpenShift Virtualization Edge	openshift-virtualization	3	edge

7. Standard Deployment Workflow

Stage	Admin/operator action
Create/import deployment	Choose a built-in template, paste/upload JSON/YAML, or define site details manually.
Select hardware	Choose generic Redfish or OEM provider and add every node with serial, BMC IP, and role.
Select platform	Choose Azure Local, vSphere, AHV, Proxmox, Hyper-V, KVM, or OpenShift Virtualization.

Validate	Checks desired-state schema and platform-specific required fields.
Inventory	Collects Redfish/BMC facts when credentials and network reachability are present.
Preflight	Combines desired state, inventory, network assumptions, and readiness checks.
Generate artifacts	Creates deployment bundle and IaC handoff outputs. Download ZIP for change record.
Request approval	Protected actions return approval-required until the required approver role/count is satisfied.
Run workflow	Run approved jobs and monitor progress logs/events. Export job report after completion.

8. Hypervisor-Specific Admin Notes

Azure Local

Template	Admin use
Azure Local Branch	Small branch/edge HCI
Azure Local Edge Scale	Four-node HA edge
Azure Local Stretched	Two-room or metro resilience

- Require subscription ID, tenant ID, resource group, region, cluster name, DNS, NTP, and VLANs.
- Keep Azure credential references in the secret provider, not in YAML notes.
- Treat deploy-azure-local as approval-gated and validate field execution per site.

VMware vSphere

Template	Admin use
vSphere Management Cluster	Core management services
vSphere Workload Cluster	Compute workload cluster

- Confirm vCenter, datacenter, datastore, cluster, and port group names before preflight.
- Use generated artifacts as handoff until provider-specific live execution is validated.
- Check DNS forward and reverse records before host onboarding.

Nutanix AHV

Template	Admin use
Nutanix AHV Edge	Prism-managed edge cluster
Nutanix AHV ROBO	Remote office profile

- Separate Prism Central and Prism Element credentials and endpoints.
- Record hardware compatibility evidence before live provider workflows.
- Document CVM, storage, and management networks explicitly.

Proxmox VE

Template	Admin use
Proxmox Lab	Compact lab
Proxmox Edge	Ceph-backed edge

- Document quorum or witness design for two-node/small clusters.
- Confirm bridge names, storage backend, repository policy, and node naming.
- Use approval gates before ISO mount or node rebuild workflows.

Hyper-V

Template	Admin use
Hyper-V Cluster	Two-node Windows virtualization
Hyper-V Workload Scale	Four-node S2D profile

- Confirm AD, DNS, NTP, domain join, virtual switch, and cluster witness design.
- Do not run cluster-impacting actions without rollback and maintenance window.
- Keep Windows/provider credentials in the secret provider.

KVM

Template	Admin use
KVM Libvirt Cluster	Linux libvirt estate
KVM OpenStack Compute	OpenStack compute foundation

- Confirm Linux distribution, repositories, storage pool, bridge names, SSH method, and automation access.
- Track drift if administrators make manual host edits between StrataOne workflows.
- Treat OpenStack-adjacent execution as provider-contract until validated in the target environment.

OpenShift Virtualization

Template	Admin use
OpenShift Virtualization	Private cloud virtualization
OpenShift Virtualization Edge	Compact edge app platform

- Confirm API endpoint, kubeconfig/credential reference, namespace, storage class, and network attachment definitions.
- Use scoped RBAC rather than cluster-admin for routine automation.
- Run storage and quota checks before VM rollout.

9. ISO Library and Virtual Media

Item	Admin detail
Host path	Repository .strataone/iso-library directory when using the supplied Compose volume.
Container path	/app/.strataone/iso-library inside strataone-api and strataone-worker containers.
Dashboard use	Artifacts > ISO Registry > Browse can select files from the configured ISO library.
Remote media	Use URI entries when ISO files are served from an approved web repository.
Live mount	mount-iso requires approval and, for real Redfish calls, STRATAONE_ENABLE_LIVE_REDFISH=true plus OEM validation.

10. Day-2 Operations

Area	Admin routine
Health	Check dashboard API badge, /health endpoint, docker compose ps, and recent job failures.
Jobs	Review orchestration runs, progress events, live logs, rerun/resume options, and job report output.
Approvals	Review pending approvals daily in active deployment periods; reject stale or unclear requests.
Audit	Review auth, role, provider, approval, and job events after each change window.

Artifacts	Download ZIP bundles and attach them to the implementation/change record.
Provider evidence	Maintain validation records before enabling live actions for a vendor/provider combination.

11. Backup and Recovery

Component	Admin action
PostgreSQL volume	Back up strataone-postgres or use native pg_dump/restore for durable state.
Redis volume	Back up if queued job state must survive maintenance; otherwise drain workers before shutdown.
.strataone directory	Back up artifacts, ISO library metadata, OEM validation evidence, and local files.
Environment file	Back up securely; it contains operational settings and may contain sensitive values.
Restore test	Regularly restore into an isolated environment and confirm login, sites, jobs, approvals, and artifacts.

12. Troubleshooting Quick Reference

Symptom	Likely admin fix
Dashboard shows anonymous/no token	Sign in through Login. If stale, logout/refresh and confirm API auth settings.
Templates unavailable	Rebuild/restart API after backend changes; /templates should return 200 without auth.
Settings HTTP 401	Session token expired or RBAC lacks permission. Sign in as Platform Administrator.
No functions work	Check browser console, API URL/proxy, auth token state, and API container health.
ISO browse empty	Create host .strataone/iso-library and restart container if the volume path was absent.
Port 3000 forbidden	Use dashboard port 8088 or override STRATAONE_DASHBOARD_PORT.
Inventory fails	Confirm BMC route, credentials, TLS/insecure setting, and Redfish endpoint behavior.
Approval required	Open Approvals, approve with required role/count, then run approved action or approve-and-run.

13. Field Readiness Checklist

Check	Pass condition	Done
Access	Named admin account works; default passwords and bootstrap tokens changed.	
Services	API, dashboard, worker, PostgreSQL, and Redis are healthy.	
Security	Auth, RBAC, CORS, rate limiting, security headers, and tenant enforcement are configured.	
Secrets	Secret provider selected; credentials stored as references.	
Approvals	Policies configured for mount-iso, deploy-azure-local, node-replacement, and other live actions.	
Providers	Provider catalog reviewed; lab-validation evidence recorded where live actions are expected.	
Templates	Built-in templates visible and import/upload path tested.	
ISO	ISO library path exists or approved remote ISO repository is configured.	
Backup	Backup and restore process tested before production change windows.	

Stop condition

If Validate or Preflight fails, stop and fix the desired state or environment. Do not use live provider execution as a discovery tool.